

Intermediate phase evolution for stable and oriented evaporated wide-bandgap perovskite solar cells

Received: 26 April 2025

Accepted: 12 September 2025

Published online: 22 October 2025

 Check for updates

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Efficient wide-bandgap perovskite solar cells have pushed tandem efficiencies to 34.9%, reinforcing their promise for next-generation photovoltaics. However, their commercial adoption is hindered by stability issues of wide-bandgap perovskites, especially under high-temperature maximum power point tracking conditions. Here we report the stabilization of ~1.7-eV wide-bandgap perovskites via intermediate phase evolution, enabling a self-guided crystal-growth mode. A CsI₂Br intermediate phase forms during early stage deposition, directing the oriented growth of polycrystalline films with unique texturing. Atomic-scale scanning transmission electron microscopy reveals that the CsI₂Br (123) facet, with a 2.9-Å interplanar spacing, matches the perovskite (200) facet, guiding coherent {100} growth. This results in enhanced crystallinity, with a 2-order-magnitude increase in the (100) diffraction intensity and a reduced full-width at half-maximum from 0.249° to 0.148°, compared with solution-processed films. The resulting solar cells exhibit outstanding thermal and operational stability, maintaining performance under maximum power point tracking for over 3,000 h at room temperature and over 500 h at 110 °C, with a projected lifetime of ~70,000 h. With 21.37% power conversion efficiency and >84% fill factor, this work presents a compelling route towards stable, high-efficiency tandem photovoltaics.

Wide-bandgap perovskites (~1.7 eV) have driven major advancements in perovskite–silicon (PK–Si) tandem solar cells, achieving a record power conversion efficiency (PCE) of 34.9% (refs. 1–4). This outstanding performance, coupled with their potential for cost-effective, scalable production and silicon’s established manufacturing infrastructure, has positioned PK–Si tandems as a promising next-generation photovoltaic technology. However, the long-term stability of such wide-bandgap perovskites remains a critical challenge^{5,6}.

Wide-bandgap perovskites are prone to degradation under accelerated aging stresses. Ji et al.⁷ reported the incorporation of an AlO_x buffer layer on the interface to stabilize the perovskite by the suppression of halide ion migration. Liu et al.⁸ replaced commonly used LiF at the perovskite/C₆₀ interface, which suffers from poor stability, with an ultrathin MgF₂ interlayer, achieving much mitigated degradation in PK–Si tandem devices under damp heat. While previous studies focused on stabilizing interfaces, enhancing

the stability of the perovskite bulk material remains a more critical yet challenging task.

For bulk perovskites, both composition and crystal quality are paramount. Regarding composition, high-Cs, methylammonium-free compositions are generally considered advantageous for stabilizing the wide-bandgap perovskite phase under harsh environmental conditions^{1,9,10}. Nevertheless, it is difficult to produce efficient devices using traditional approaches due to the low solubility of high-Cs compositions and high defect density in methylammonium-free perovskites. In terms of crystal quality, perovskite crystals that are poorly oriented or defect rich are more vulnerable to degradation, emphasizing the need to enhance the bulk crystallographic quality for both high efficiency and long-term stability in solar cells^{11–13}.

Studies have shown that specific crystal facets substantially influence the photovoltaic performance and stability. For instance, experimental and theoretical work by Ma et al. reported that the photovoltaic properties of different facets follow the order (100) $\geq$ (111) $\gg$ (110) (ref. 14). Devices with a preferential (100) orientation have already achieved efficiencies exceeding 25% (refs. 15–20). Recent perspectives also suggest that high-Miller-index facets may improve stability by enhancing grain structure and reducing grain boundaries²¹. These findings underscore the importance of controlling the crystal orientation in perovskite films. However, achieving both high efficiency and long-term stability remains challenging in wide-bandgap perovskites. To meet the operational warranty standard as silicon photovoltaics, substantial advancements in the intrinsic bulk quality of perovskite films are imperative.

Here we demonstrate the intermediate phase evolution of wide-bandgap perovskites via self-guided crystal growth (SCG) during the thermal evaporation deposition process. At an early stage of deposition, spontaneously formed CsI₂Br, as an intermediate phase, directed the oriented growth of a polycrystalline thin film with unique texturing. Atomic-scale scanning transmission electron microscopy (STEM) confirmed that the CsI₂Br (123) facet aligns well with the perovskite (200) facet, guiding coherent growth along the {100} facets. This resulted in particularly improved crystallinity along the (100) facet and a reduced full-width at half-maximum (FWHM) of the diffraction peak. The resulting perovskite shows mitigated formamidinium bromide (FABr) loss under thermal stress, preventing surface vacancies and slowing degradation. As a result, the thermally evaporated wide-bandgap perovskite solar cells (PSCs) maintain stable power output for over 3,000 h at room temperature, and achieve a T_{80} lifetime exceeding 538 h at 110 °C. Accelerated aging tests across temperatures from 65 to 110 °C project operational lifetimes converging at approximately 70,000 h, consistent with the T_{80} lifetime of 70,191 h derived from room-temperature measurements. Furthermore, these results indicate substantial power supply when operated in various environments, including Singapore, China and some of the sunniest regions of the United States.

SCG enabled by intermediate phase evolution

Compared with common solution processing for perovskite deposition, a vapour-based method has not been intensively explored until very recently. However, it has showcased some undeniable advantages to ensure precise thickness control, conformal film uniformity and large-area compatibility^{22–24}. In this study, we developed a co-evaporation method with precursors of caesium bromide (CsBr), formamidinium iodide (FAI) and lead(II) iodide (PbI₂). The composition of the as-deposited perovskite is Cs_{0.45}FA_{0.55}Pb(I_{0.85}Br_{0.15})₃, confirmed by energy-dispersive X-ray spectroscopy (EDX) (Supplementary Fig. 1 and Supplementary Table 1). Correspondingly, a wide bandgap of 1.69 eV is achieved, confirmed by ultraviolet–visible (UV–vis) absorption spectra (Supplementary Fig. 2). We prepared spin-coated perovskites as a reference with an identical bandgap (Supplementary Fig. 2) based on the same precursors (Methods). Indium tin oxide (ITO) substrates

are used for thin-film deposition as well as device fabrication of wide-bandgap PSCs.

Despite limited research in the past, it has been proven that controlling the microstructure of thermally evaporated perovskite films is more complicated due to fundamentally different crystallization kinetics^{22,25–27}. On the basis of our characterizations, the structure and morphology of our evaporated perovskites are distinct from the solution-processed reference, indicating the underlying difference in perovskite nucleation and grain growth mechanisms. Compared with the typical morphology of clear grain boundaries for solution-processed films, the evaporated perovskite films exhibit homogeneously packed morphology with a smooth and flat surface (Supplementary Fig. 3). More impressively, the X-ray diffraction (XRD) pattern of the evaporated films show strong (100) and (200) facet orientations with much-enhanced intensity (two-order magnitude increase in main peaks), whereas a random orientation is observed in the reference films (Fig. 1a,b). Supplementary Fig. 4 shows that crystal growth in the reference films show random orientation with a peak intensity ratio of the (100)/(110) facet of 0.69:1, whereas the evaporated films exhibit a sharp contrast in a ratio of 252:1. Moreover, a greatly narrowed FWHM of 0.148° for (100) peak is obtained compared with 0.249° of the reference, showing much improved crystallinity and crystal quality. To further visualize the facet orientation difference, pole figure measurements were implemented (Fig. 1c,d and Supplementary Fig. 5), in which the much more concentrated signal at the centre along (100) peak suggests a far stronger preferred orientation of the films perpendicular to the substrate^{28,29}. In particular, the concentrated signal of the evaporated perovskites shows a high intensity, and the degree of the (100) preferred orientation was well correlated with the XRD patterns shown in Fig. 1a,b. Moreover, the signal is highly concentrated to a spot-like region, meaning that it is fully ordered both along the surface normal and in the sample plane, similar to epitaxial films grown on single-crystal substrates^{16,18,19,28}.

In vapour-based deposition, the precursors are usually sublimed by heating to specific temperatures, interacting with each other and condensed when reaching the cold substrate. However, the vapour deposition of perovskite materials tends to be more complicated than a simple physical process, especially for co-evaporation based on more than one precursor^{30,31}. To reveal the underlying crystal growth mechanism during the evaporation process, we conducted thickness-dependent STEM for perovskite films with thicknesses from 20 to 100 nm. The high-angle annular dark-field (HAADF) STEM images and EDX mapping results (Supplementary Fig. 6) show the evolution of morphology and composition from the start of deposition. Different from the common understanding that the perovskite is directly condensed onto the substrate, an intermediate phase is observed to be spontaneously formed in the films with a composition rich in Cs, I and Br, surrounded by some primary perovskite crystals with small grains. To further determine the identity of this secondary phase and the role it plays, we use a series of low-dose atomic-scale high-resolution STEM measurements, which can provide fine structural details of electron-beam-sensitive perovskite materials. First, the high-resolution STEM image shown in Fig. 1f together with the corresponding fast Fourier transform (FFT) pattern shown in Fig. 1g reveal the crystal structure and atomic arrangement of the as-formed secondary phase (Supplementary Fig. 6, red circle) at the early stage of deposition. After careful lattice calibration and resolution, this structure is indexed to CsI₂Br in the orthorhombic space group of *Pnma* (number 62) with the unit-cell parameters of $a = 6.75$ Å, $b = 9.85$ Å and $c = 10.93$ Å (refs. 32,33). More detailed structural information about CsI₂Br is provided in Supplementary Fig. 7. To figure out the formation pathway of CsI₂Br from the precursors at the early stage of evaporation, we conducted mass spectroscopy measurements at a high temperature for the precursor materials to determine the generated products in the vapour phase. For the PbI₂ precursor, when heated

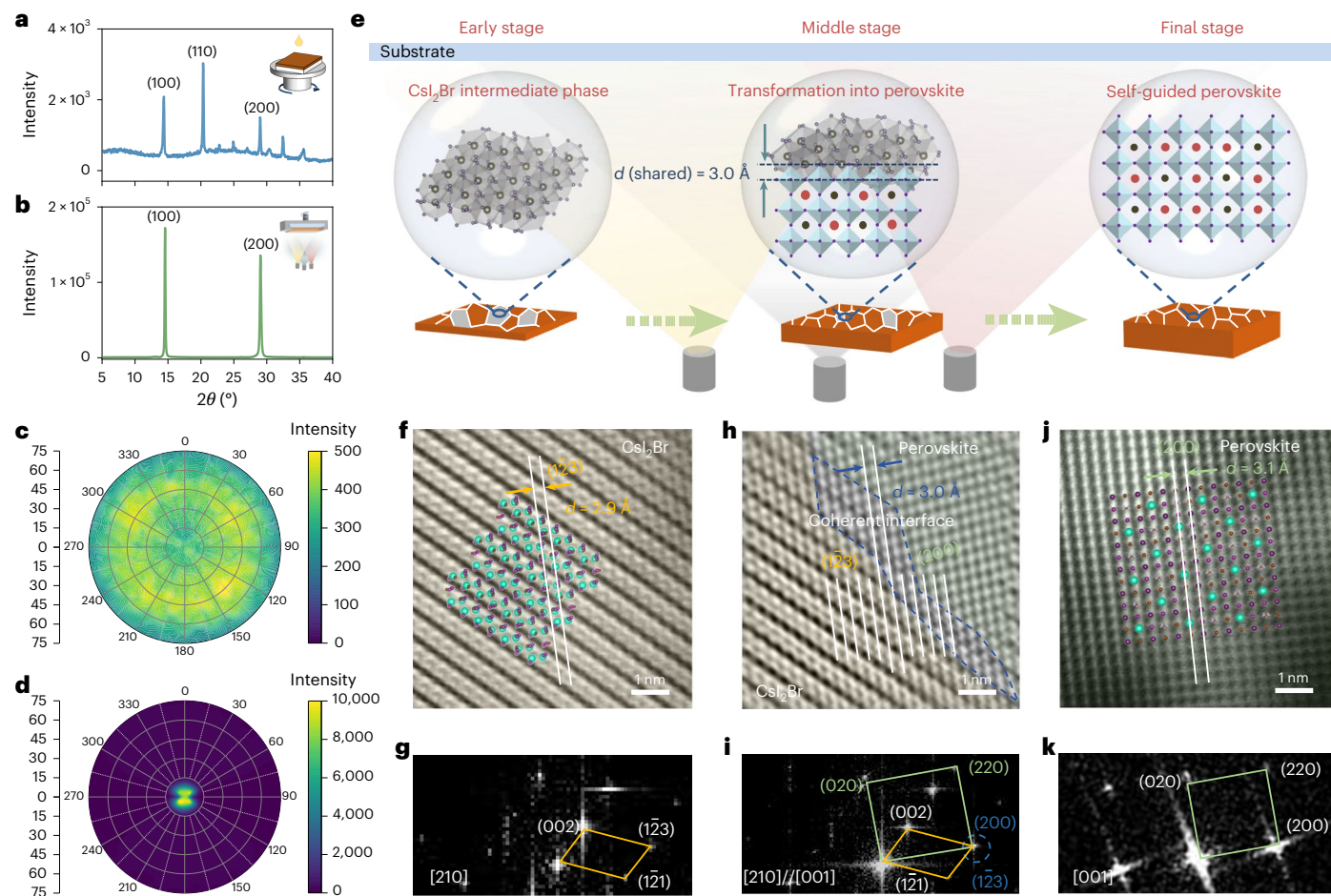


Fig. 1 | Atomic-scale high-resolution STEM images and SCG induced by CsI_2Br intermediate phase. **a, b**, XRD patterns of perovskite thin films without (**a**) and with (**b**) SCG. Inset: illustrations of the deposition methods of perovskite thin films without and with SCG, respectively. **c, d**, Pole figures of perovskite thin films from the (100) facet without (**c**) and with (**d**) SCG. **e**, Schematic of CsI_2Br -induced SCG of perovskite during the deposition process. **f, g**, High-resolution STEM image (**f**) and the corresponding FFT patterns (**g**) of an inorganic CsI_2Br

intermediate phase. The theoretical atomic model in the STEM image shows the corresponding atomic positions of CsI_2Br . **h, i**, High-resolution STEM image (**h**) and the corresponding FFT patterns (**i**) of transformation between CsI_2Br and perovskite. **j, k**, High-resolution STEM image (**j**) and the corresponding FFT patterns (**k**) of a well-developed perovskite with SCG. The theoretical atomic model in the STEM image shows the corresponding atomic positions of perovskite crystals.

at 350 °C, Supplementary Fig. 8 shows that it was decomposed into gas-state I_2 and other lead-containing vapours including Pb and PbI . For the CsBr precursor, due to the higher bond dissociation energy of Cs-Br (ref. 34), only the signal of gas-state CsBr was observed when heated at 400 °C. Therefore, we infer that the formation of CsI_2Br is coming from the reaction between gas-state CsBr and I or I_2 at a high temperature under vacuum.

We then investigate its relationship with perovskite crystals and how it influences the crystal growth of the perovskite. When zooming into a nanoscale scope between the perovskite crystal and CsI_2Br phase, transition zones along the interface contact are observed. Figure 1h,i reveals the coherent interface formed between the (123) facet of CsI_2Br and (200) facet of perovskite with their well-aligned interplanar spacing of 2.9 Å and 3.1 Å, respectively. The shared facet of these two structures shows an interplanar spacing of 3.0 Å due to some acceptable strains at the heterointerface. The coherent growth between CsI_2Br and perovskite indicates the succession of the (200) facet orientation in perovskite from the (123) facet in CsI_2Br , meaning that the first formed CsI_2Br acts as an intermediate phase for perovskite formation and guides the crystal growth of perovskites through the coherent interface. Finally, we characterized the well-developed perovskite crystals after complete deposition, and Fig. 1j,k shows that the as-deposited perovskites maintain the preferential orientation along the {100} facets.

At this point, the crystal growth mechanism can be visualized and summarized into three stages (Fig. 1e). At the early stage of evaporation, an intermediate phase of CsI_2Br is first formed spontaneously in the films with the (123) facet exposed, the d spacing well matches with the (200) facet of perovskites. At the middle stage, as evaporation proceeds, the crystal growth of the perovskite is guided by the (123) facet of CsI_2Br , leading to oriented crystallization along the {100} facets with succession from the coherent interface. At the final stage, the well-developed perovskite crystals maintain the preferential orientation, finally forming the perovskite film showing high orientation. To highlight this unique mechanism, we denote it as perovskites with SCG throughout this work.

Structural stability of perovskite under thermal stress

The degradation of organic–inorganic halide perovskites at elevated temperatures due to vapourization and loss of various organic species has been widely reported^{1,35–40}. Specifically, for $\text{Cs}_x\text{FA}_{1-x}\text{PbI}_{3-y}$ perovskites, the loss of organic ions was demonstrated to happen even under mild annealing at as low as 100 °C (refs. 40,41). The FA^+ loss has been demonstrated to be easier to happen with the co-existence of Br , which lowers the vacancy formation energy when exposed to a high temperature³⁹.

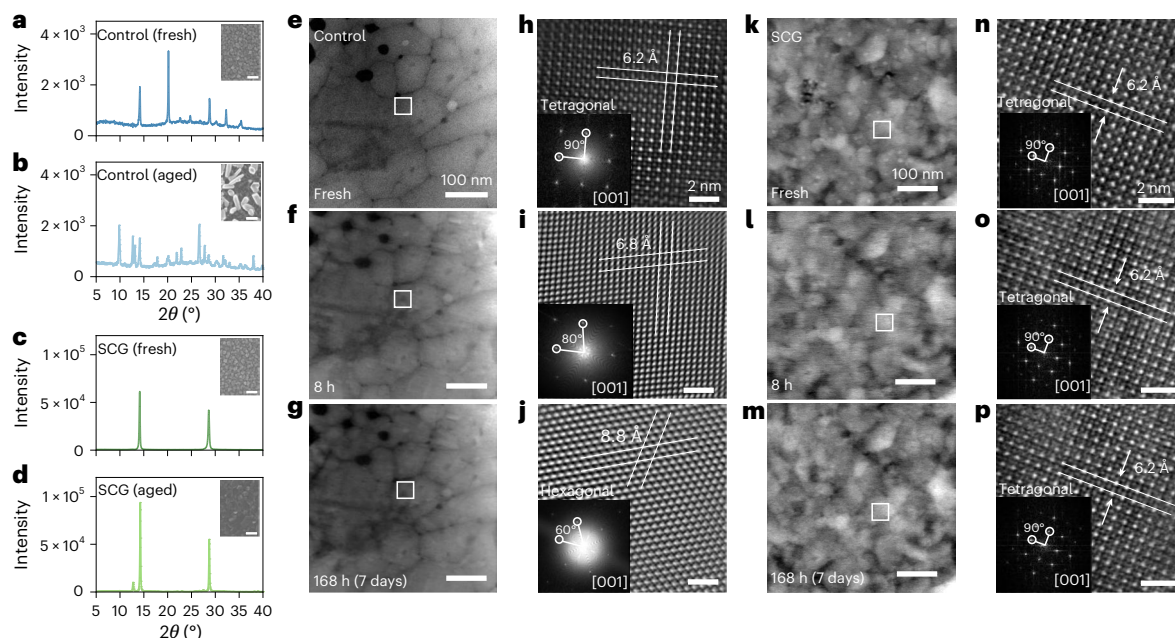


Fig. 2 | Structural evolution of perovskites under thermal stress revealed by STEM. **a, b**, XRD patterns of control perovskite thin films without SCG before **(a)** and after **(b)** aging at 110 °C for 3 days. Inset: SEM images of the corresponding thin films. Scale bars, 500 nm. **c, d**, XRD patterns of perovskite thin films with SCG before **(c)** and after **(d)** aging at 110 °C for 15 days. Inset: SEM images of the corresponding thin films. Scale bars, 500 nm. **e–g**, HAADF-STEM images of the control perovskite thin films without SCG before aging **(e)**, after 8 h

of continuous aging **(f)** and after 7 days of continuous aging **(g)** at 110 °C. **h–j**, Corresponding STEM images zoomed in from the squares in **e–g** (**h–j**, respectively). Insets: the corresponding FFT patterns. **k–m**, HAADF-STEM images of perovskite thin films with SCG before aging **(k)**, after 8 h of continuous aging **(l)** and 7 days of continuous aging **(m)** at 110 °C. **n–p**, Corresponding STEM images zoomed in from the squares in **k–m** (**n–p**, respectively). Insets: the corresponding FFT patterns.

On the basis of these features of $\text{Cs}_x\text{FA}_{1-x}\text{PbI}_{3-y}\text{Br}_{3-y}$ perovskites, we characterize the perovskite films with and without SCG under long-term thermal stress at an elevated temperature of 110 °C, simulating the most extreme condition PSCs may face during their real-world operation. The control perovskite films undergo fast degradation from perovskite into undesirable phases within a short time of 3 days (Fig. 2a,b and Supplementary Fig. 9). The obvious drops in the XRD peaks of the perovskite together with the rising peaks of PbI_2 and other phases indicate the structural deterioration of perovskite crystals under thermal stress. In sharp contrast, for the perovskite films with SCG, the high XRD intensity of the perovskite is well maintained and even enhanced after aging at 110 °C for more than 15 days (Fig. 2c,d and Supplementary Fig. 9). In Supplementary Figs. 10 and 11, UV–vis absorption and steady-state photoluminescence measurements reveal that the control films are no more effective active layers after high-temperature aging, whereas the films with SCG maintain their optical functionality for more than 15 days.

To further investigate the thermal degradation pathway, STEM is used to probe the microscopic structural evolutions. First, the high-resolution STEM images from Fig. 2e–g illustrate the morphology change in the control perovskite under thermal stress. An undesirable phase was observed in the film with rough and inhomogeneous morphology as high-temperature aging proceeds. Before high-temperature aging, the control perovskite exhibits a tetragonal or cubic structure with an interfacial angle of 90°, and the interplanar spacing of (100) planes are about 6.2 Å (Fig. 2h and Supplementary Fig. 12). After aging at 110 °C for 8 h, the interfacial angle changes to 80° with an enlarged d spacing into 6.8 Å, suggesting an intermediate transition stage with some lattice distortion (Fig. 2i and Supplementary Fig. 12). After further aging for 7 days, the interfacial angle becomes 60° and the d spacing changes to 8.8 Å, indicating the final transition into hexagonal PbI_2 (Fig. 2j and Supplementary Figs. 12 and 13). This observed degradation from

perovskite to PbI_2 is consistent with the XRD results. By contrast, as shown in Fig. 2k–m, there is no obvious change in morphology during the aging period. In the zoomed-in views, we found that it maintains the same structure with an interfacial angle at 90° and interplanar spacing of (100) crystal planes at 6.2 Å (Supplementary Fig. 14), indicating negligible distortion of the crystal structure under harsh thermal stress.

Revealing facet-dependent thermal degradation pathway

On the basis of the observations above, we further conduct elemental analysis during high-temperature aging to characterize the compositional evolution. First, based on the EDX mappings in Supplementary Figs. 15 and 16, we extracted the relative ratios of Br:Pb (Fig. 3a), Cs:Pb and I:Pb (Supplementary Fig. 17) at different aging times. Figure 3a presents that the Br:Pb ratio in control perovskite film decreases constantly and drops to 68% of the initial value after 7-day aging, indicating a severe Br loss under thermal stress. However, for the target perovskite film, such a decreasing trend is mostly mitigated and 98% of the initial Br:Pb ratio can be retained after 7 days of aging at 110 °C. For Cs and I elements, their relative ratio with Pb varies within 8% range of their initial values in both films with and without SCG, indicating negligible compositional changes in Cs and I. For organic compositions of perovskites (which are difficult to analyse using EDX), X-ray photoelectron spectroscopy (XPS) is used for further elemental analysis. Figure 3b,c shows the N1s peaks from the FA^+ cation of control and target perovskites before and after thermal aging, respectively. The integrated peak area for the control after aging decreases to only 47.78% of the initial area before aging. For the target sample, 90.44% of the initial peak area is maintained after aging for 7 days. This indicates that FA^+ loss occurs at the surface of the control films under thermal stress. As shown in Supplementary Fig. 18, the same Br loss trend is also observed from the integrated areas of the Br3d peaks

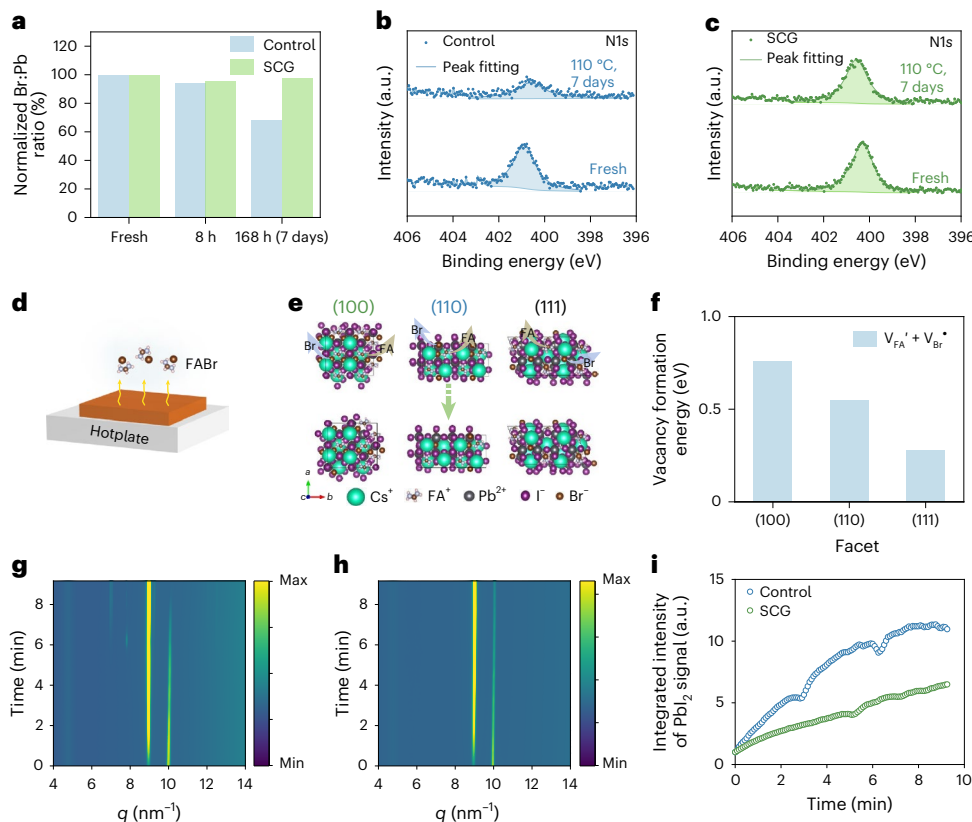


Fig. 3 | Revealing facet-dependent thermal degradation pathway. **a**, Evolution of normalized Br:Pb ratios determined from STEM-EDX mappings of perovskite thin films without and with SCG under 110 °C aging for 0 h (fresh), 8 h and 168 h. **b,c**, Comparisons of XPS N1s signals between fresh and aged (at 110 °C for 7 days) perovskite thin films without (**b**) and with (**c**) SCG. The solid lines represent the fitting of XPS data. **d**, Schematic of FABr loss during the thermal aging of perovskite thin films on a hotplate. **e**, DFT-simulated crystal structures of perovskites sliced along the (100), (110) and (111) facets before (upper row)

and after (lower row) the formation of V_{FA}' and V_{Br}' . **f**, Vacancy formation energies calculated from the DFT simulation for the formation of V_{FA}' and V_{Br}' for the (100), (110) and (111) facets of the perovskite. **g,h**, In situ GIWAXS measurements under a high temperature of 240 °C on perovskite films without (**g**) and with (**h**) SCG. **i**, Comparison of the integrated PbI_2 peak intensity of perovskite films without and with SCG extracted from **g** and **h**. The full azimuthal range of the two-dimensional image was integrated for the integrated signals.

for the control perovskite, considering the negligible compositional change in the other elements. On the basis of this elemental analysis near the surface, we confirm that the thermal degradation of control perovskites starts from the compositional losses of FA^+ and Br^- under thermal stress. With SCG, the compositional loss is notably mitigated, leading to much-enhanced structural stability against heat. On the basis of the compositional analysis above, we point out that there is a FABr loss process under thermal stress (Fig. 3d).

Diving into mechanisms behind the mitigated compositional loss by the SCG process, we performed density functional theory (DFT) calculations. We established the models as illustrated in Fig. 3e and detailed in Supplementary Note 1. The calculation results show that the vacancy formation energy of a pair of V_{FA}' (FA^+ vacancy) and V_{Br}' (Br^- vacancy), which are associated with FABr loss, varies with different facets and Fig. 3f shows that it arranges in the following order: (100) > (110) > (111). The highest vacancy formation energy for the (100) facet suggests that the formation of V_{FA}' and V_{Br}' is the most difficult thermodynamically at the perovskite surface with the (100) facet exposed, suppressing FABr loss from the (100) facet. However, in the control sample, the rapid crystallization of the perovskite leads to a random orientation, resulting in a relatively high proportion of (110) crystal facets exposed on the surface, causing FABr loss and subsequent degradation. By contrast, for the target perovskite with SCG, the crystal orientation is effectively tailored to the (100) facet by coherent growth from the CsI_2Br intermediate phase, fundamentally suppressing the formation of FA^+ and Br^- vacancies on the surface to mitigate thermal degradation initiated by FABr loss.

To obtain a more intuitive view of thermal degradation, we applied an extremely high temperature of 240 °C to the perovskite films and conducted in situ grazing-incidence wide-angle X-ray scattering (GIWAXS) characterization (Fig. 3g,h). Under such harsh conditions, the degradation of both control and target perovskite films are considerably accelerated, as evidenced by the decreased Bragg (200) peak of perovskites. However, when we quantify the degree of PbI_2 formation by integrating the peak intensity at different times, the control sample shows a much faster degradation than the perovskite with SCG (Fig. 3i). Meanwhile, undesirable phases including δ - $CsPbI_2Br_{3-z}$ appear in the control film after 6 min of heating, indicating severe degradation happening in the control sample under thermal stress^{42,43}. Further, we quantify the phase transformation kinetics with data extracted from the GIWAXS results, using the Johnson–Mehl–Avrami–Kolmogorov equation⁴⁴ (Supplementary Note 2). As shown in Supplementary Fig. 19 and Supplementary Table 2, the perovskite films with SCG exhibit suppressed thermal degradation with a smaller transformation rate constant K of $7.3 \times 10^{-5} s^{-1}$ compared with $1.1 \times 10^{-4} s^{-1}$ for the control perovskite films.

Projecting lifetime of perovskites with SCG under various environments

Leveraging the aforementioned benefits, we integrated the optimized perovskite with SCG into p–i–n PSCs (Supplementary Fig. 20). Figure 4a displays the champion current–voltage (J – V) characteristics of the devices, whereas Supplementary Table 3 summarizes their

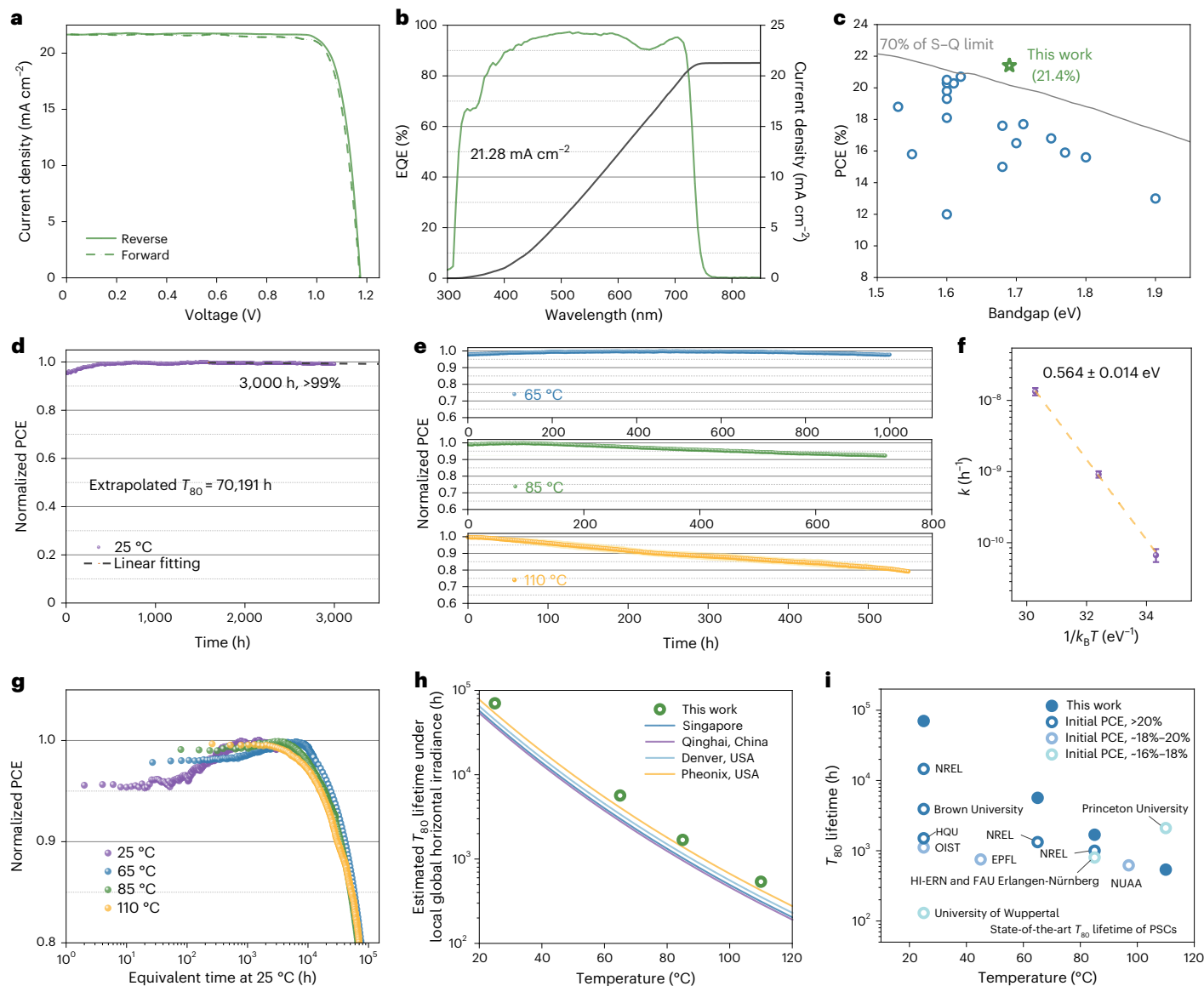


Fig. 4 | Projecting operational lifetime of perovskites with SCG in different environments. **a**, Champion $J-V$ curves of PSCs with SCG in forward and reverse scans. **b**, EQE curves and integrated J_{sc} values for PSC with SCG. **c**, Comparisons between champion PSC with SCG of this work and previous reports of evaporated PSCs. The solid line shows 70% of the Shockley–Queisser (S–Q) limit of the PCE value. **d**, Measured MPPT stability of PSCs with SCG in terms of normalized PCE at 25 °C and the linear extrapolation of the measured evolution of normalized PCE. The initial device PCE is 21.35%. **e**, Measured stability of PSCs with SCG in terms of normalized PCE at 65 °C, 85 °C and 110 °C. The data points are averaged from three independent devices for each temperature. The initial PCEs measured at room temperature of these devices for MPPT tests at different temperatures can be found in Supplementary Table 6. The coloured areas near the curves represent standard deviations of the normalized PCE distributions

and data are presented as mean values $\pm$ standard deviation. **f**, Extracting E_a from the calculated degradation rate k and $1/k_B T$ from measurements at 65 °C, 85 °C and 110 °C. The error bars represent the standard deviations of the three calculated degradation rates and data are presented as mean values $\pm$ standard deviation. **g**, Equivalent MPPT stability at different temperatures derived from multiplying the original MPPT results represented by solid circles (with linear extrapolation represented by hollow circles in the same color) with the specific acceleration factor for that temperature. **h**, Comparison between equivalent T_{80} lifetimes of PSC with SCG of this work at different temperatures (dots) with the minimum requirements for the lifetime at 25 °C to reach 25 years (lines) at 4 representative locations. **i**, Equivalent T_{80} lifetimes of PSC with SCG of this work with the reported T_{80} lifetimes at different temperatures. The references are listed in Supplementary Table 6.

photovoltaic parameters. The perovskite with SCG delivers a PCE of 21.37% with a fill factor exceeding 84%. External quantum efficiency (EQE) measurements (Fig. 4b) show well-performing photovoltaic response for perovskite with SCG, leading to an integrated short-circuit current density of 21.28 mA cm⁻². The determined photovoltaic bandgap (Supplementary Fig. 21) aligns well with the optical bandgaps presented in Supplementary Fig. 2. We included 12 devices in the same batch for reproducibility (Supplementary Table 4 summarizes their performances). In particular, compared with previously reported PSCs based on evaporated perovskites, our devices, to the best of our knowledge, deliver the highest

efficiency regardless of the bandgap (Fig. 4c and Supplementary Table 5). This is a milestone for the vacuum-based processing of perovskites, especially for one-step co-evaporation. Moreover, bias-assisted charge extraction measurements⁴⁵ show that the evaporated devices with SCG exhibit a reduced mobile ion concentration compared with the control (Supplementary Fig. 22). Considering the importance of ion-induced degradation losses to the operational stability⁴⁵, we expect that this will be beneficial for achieving longer operational lifetimes.

Considering the excellent thermal stability of our perovskite with SCG, we conducted a series of standard stability tests based on

International Summit on Organic Photovoltaic Stability (ISOS) protocols⁴⁶. For the ISOS-D-1 test (Supplementary Fig. 23), the devices show no performance decrease after more than 4,000 h. Maximum power point tracking (MPPT) is further conducted in the testing room with a high relative humidity of 85% (harsher than the ISOS-L protocols). At the ambient temperature, Fig. 4d shows that the devices can maintain more than 99% of the highest efficiency after 3,000 h, showing an extrapolated T_{80} lifetime of 70,191 h. We further conducted accelerated aging experiments at elevated temperatures ranging from 65 to 110 °C for reliable lifetime prediction^{46–49}. The evolution of normalized PCEs is depicted in Fig. 4e and Supplementary Fig. 24, showing T_{80} lifetimes from 5,657 h at 65 °C to 538 h at 110 °C. The degradation rates from accelerated aging tests show a linear decay behaviour as

$$k = -\frac{d\text{PCE}(t)}{dt},$$

where k is the degradation rate and t is time. The data in Fig. 4f are plotted according to the Arrhenius model^{47,48,50,51} (Supplementary Note 3) and an apparent activation energy (E_a) of approximately 0.56 eV can be extracted, which is comparable with the reported values in recent studies^{47,52–54}. To project the operational lifetime at room temperature, the acceleration factors are calculated in Supplementary Fig. 25. As shown in Fig. 4g, the projected operational lifetimes of different temperatures converge at around 70,000 h, perfectly aligning with the extrapolated T_{80} lifetime from room-temperature measurement. Specifically, based on the measured T_{80} lifetime of 538 h at 110 °C, the estimated T_{80} lifetime at 25 °C can be calculated as 70,070 h with an acceleration factor of 130.

Furthermore, the operation lifespan is influenced by specific local irradiation. We calculated the required T_{80} lifetime in various locations as a function of temperature for theoretical 25-year operation (Fig. 4h), and the relationship is detailed in Supplementary Note 4 (refs.^{55,56}). All the green dots representing the measured data in this work are above the estimated lines at various temperatures, indicating the potential of PSCs with SCG to deliver a substantial power supply in these locations. Additionally, compared with previously reported T_{80} lifetimes at different temperatures (Fig. 4i and Supplementary Table 6), the lifetime of the optimized devices rival the best-to-date values at various elevated temperatures, especially among the wide-bandgap PSCs (Supplementary Fig. 26).

The excellent performance of wide-bandgap perovskites also promises the massive potential of PK–Si tandems. Consequently, we prepared PK–Si tandem solar cells based on the perovskite with SCG by evaporation (Supplementary Fig. 27). The tandem devices show a champion PCE of 29.43% for an area of 1 cm², demonstrating the smooth transfer and universal application of the SCG strategy by evaporation from a single junction to a tandem configuration.

Online content

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Methods

Materials

CsBr (99.999%), PbI_2 (99.99%), tetrakis-(dimethylamino) tin(IV) (99.99%), *N,N*-dimethylformamide (anhydrous), dimethyl sulfoxide (anhydrous) and 2-propanol (IPA; anhydrous) were purchased from Sigma-Aldrich. FAI was purchased from Greatcell Solar Materials. [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid was purchased from Tokyo Chemical Industry. Silver pellets were purchased from Alfa Aesar. C_{60} (99.9%) was purchased from Lumtec. Nickel oxide (NiO_x) nanoparticles were synthesized following a previous report⁵⁷.

Wide-bandgap perovskite single-junction device fabrication

Patterned ITO glass substrates were cleaned with a surfactant solution and then soaked in deionized water, acetone and IPA in an ultrasonic bath for 15 min each. After UV–ozone treatment for 15 min, the substrates were then spin coated with NiO_x (10 mg ml^{-1} in deionized water) nanoparticle dispersion. The NiO_x films were annealed at 120 °C for 10 min. Then, the films were transferred to a nitrogen-filled glovebox and [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (1 mg ml^{-1} in IPA) was spin coated on the substrates at 4,000 rpm for 30 s and heated at 100 °C for 10 min.

For control perovskites based on solution method: the 1 M perovskite ($\text{Cs}_{0.45}\text{FA}_{0.55}\text{Pb}(\text{Br}_{0.15}\text{I}_{0.85})_3$) precursor solution used contains FAI, CsBr and PbI_2 that were dissolved in 1 ml mixed solvent of *N,N*-dimethylformamide:dimethyl sulfoxide (v/v = 4:1). For the spin-coating process, the substrate was spun at 4,000 rpm for 60 s with an acceleration of 4,000 rpm s^{-1} and nitrogen gas was blown on top of the spinning substrates after 20 s. The perovskite films were then annealed at 100 °C for 15 min. The as-deposited perovskite films were transferred to the thermal evaporator where 20 nm of C_{60} was deposited. Thereafter, 10 nm of SnO_2 buffer layer was deposited in an atomic layer deposition system using tetrakis-(dimethylamino) tin(IV) and H_2O . Finally, Ag was deposited as top contact to complete the device fabrication.

For SCG perovskites based on evaporation: the perovskite material was evaporated with PbI_2 , CsBr and FAI as precursors using the thermal evaporation tool in a nitrogen glovebox. The deposition rates were monitored using quartz crystal microbalance sensors. The measured rates were used to adjust the source temperature dynamically during the process to maintain a constant rate. The precursor deposition rates were 2 $\text{\AA s}^{-1}$ for FAI, 2 $\text{\AA s}^{-1}$ for PbI_2 and 0.9 $\text{\AA s}^{-1}$ for CsBr. During deposition, the base pressure was maintained at around 1×10^{-5} mbar. The chamber and substrate temperature were set to 20 °C and the rotation speed of the substrate holder to 10 rpm. The deposition was finished when the film thickness reached 600 nm. The total processing time for perovskite deposition is around 50 min. Without further annealing, the as-deposited perovskite films were transferred to the thermal evaporator where 20 nm of C_{60} was deposited. The subsequent fabrication procedure is the same as above.

Perovskite–silicon tandem device fabrication

For perovskite–Si tandem devices, the TOPCon silicon bottom cells with ITO were first treated by UV–ozone for 10 min, and the substrates were then spin coated with NiO_x (10 mg ml^{-1} in deionized water) nanoparticle dispersion. The NiO_x films were annealed at 120 °C for 10 min. Then, the films were transferred to a nitrogen-filled glovebox and [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (1 mg ml^{-1} in IPA) was spin coated on the substrates at 4,000 rpm for 30 s and heated at 100 °C for 10 min. The perovskite layer was then co-evaporated with PbI_2 , CsBr and FAI as precursors during thermal evaporation with the same process as for single-junction devices. The deposition was finished when the film thickness reached 600 nm. Without further annealing, the as-formed perovskite films were transferred to the nitrogen-filled glovebox. After that, 15 nm of C_{60} was deposited by the thermal evaporator, 10 nm of SnO_2 was deposited by the atomic layer deposition system and 80 nm

of ITO was deposited by the magnetron sputtering system. Finally, 350 nm of silver frame and 100 nm of LiF were deposited by the thermal evaporator to finish the tandem device fabrication.

Device characterizations

For single-junction devices, the completed devices were masked with metal aperture masks (0.055 cm^2) for J – V measurement by a Keithley 2400 source meter under simulated 1-sun AM1.5 G illumination (100 mW cm^{-2}) with an ABET Technologies Sun 2000 solar simulator in a nitrogen-filled glovebox at 20 °C. The step size of the J – V scans was set to 20 mV with an integration time of 10 ms and a dwell time of 10 ms. For tandem devices, the J – V measurements of the perovskite–Si tandem solar cells were recorded with SINUS-220 (WAVELABS) under simulated 1-sun AM1.5 G illumination (100 mW cm^{-2}) in air, with a scanning rate of 100 mV s^{-1} (voltage step of 10 mV and delay time of 200 ms). The active area was determined by the aperture shade mask with an area of 1 cm^2 placed on top of the tandem solar cells.

The stability tests of the devices were conducted according to the ISOS-L-3 protocol by the MPP Tracking-4B source measure unit system (Shenzhen Lancheng Technology) under 100 mW cm^{-2} white light-emitting diode in an ambient atmosphere at a temperature of 25 °C and ~85% relative humidity in the testing room with encapsulation. For accelerated aging tests with elevated temperatures, a temperature control attachment was added to the measure unit system. For stability tests, Me-4PACz was replaced by poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) and Ag was replaced by Au. The devices were encapsulated in a nitrogen-filled glovebox by blanket sealing (UV-curable epoxy, Avantama). The cover glass was attached to the side with electrode of the device with drops of encapsulation adhesive and the adhesive is then cured with UV light for a few seconds. EQE measurements of single-junction devices were conducted using a Bentham PVE300-IVT system. The intensity of the light-emitting diode laser used was calibrated with built-in silicon and germanium diodes before measurements. These EQE measurements were conducted in an ambient atmosphere with a temperature of 25 °C and ~60% relative humidity with encapsulation. EQE for tandem devices was conducted with the Quantum Efficiency Measurement System with voltage bias and multiple light-emitting diode source bias (Enlitech). The 450-nm bias illumination was used to saturate the top cell and the 850-nm bias illumination was used to saturate the bottom cell. Cross-sectional SEM images of the devices were taken with a Regulus SU8200 system (Hitachi) at a 5-kV accelerating voltage under the secondary-electron mode.

The bias-assisted charge extraction measurement is performed under dark, in which the device was initially held at a voltage close to the open-circuit voltage, where the injected charge equals the short-circuit current. After a preset delay time, a bias of 0 V was applied to extract the injected and capacitive charges in the device. The delay times for the devices were chosen to be typically five times longer than the extraction time of charges observed under the collection bias (typically ~5–10 s) to allow ionic charges to distribute throughout the active layer. A voltage close to the V_{oc} was applied to the cells to prevent a premature extraction of mobile ions to the transport layers by the built-in voltage. However, the built-in voltage remains roughly compensated by the applied voltage, keeping the mobile ions roughly distributed throughout the bulk. Finally, after ~10 s, the voltage was switched to 0 V to extract the charge carriers. Finally, the extracted charge was obtained by integrating the current transient and the charge carrier density by dividing the total charge by the elementary charge and cell volume.

Material characterizations

UV–vis absorbance spectra were recorded using an Agilent Cary 7000 spectrophotometer. The lamp was preheated for 30 min before conducting the measurements. XRD patterns were recorded on a Bruker D8 Advance X-ray diffractometer with Cu K α radiation ($\lambda = 0.1542$ nm). Pole figure measurement was performed using the Rigaku SmartLab

3-kW device. XPS measurements were performed using the Kratos Axis Ultra XPS system. All the XPS results are calibrated by fixing the C1s peak at 284.8 eV. Top-view SEM images of devices were taken using a Regulus SU8200 system (Hitachi) at 5-kV or 3-kV accelerating voltage under the secondary-electron mode.

Synchrotron-based *in situ* GIWAXS measurements were recorded at the NCD-SWEET beamline at the ALBA Synchrotron Light Source (Spain) using a monochromatic X-ray beam with an energy of 12.95 keV and a beam size of $150 \times 25 \mu\text{m}^2$. The scattered signals were recorded using a Rayonix LX255-HS area detector, which was placed at 241.1 mm from the sample position. All samples were measured under a nitrogen-filled closed chamber and the temperature was controlled via a Linkam stage modified for grazing measurements. The grazing-incidence angle is 1° for all films in the GIWAXS measurements. The reciprocal space $|q|$ was calibrated using Cr_2O_3 from NIST as the calibration standard, obtaining a sample-to-detector distance of 210 mm. Data were processed using a custom-made Python routine based on PyFAI⁵⁸.

Transmission electron microscopy

For quasi-*in situ* transmission electron microscopy analysis, coordinate Cu grids were used to allow the precise localization and tracking of structural changes. Films were deposited directly onto the Cu grids and subjected to heating in an inert nitrogen environment. This approach facilitated transmission electron microscopy investigations of the same sample regions under thermal aging for different times, enabling a detailed analysis of structural evolution and preserving spatial consistency. This methodology provided critical insights into the stability of perovskite materials during the thermal aging process.

HAADF-STEM imaging was performed using spherical-aberration-corrected microscopes, including Titan-G2 and JEOL ARM-300 operating at 300 kV. HAADF-STEM imaging, conducted with inner and outer collection angles of 58.5 mrad and 200 mrad, respectively, achieved sub-ångström resolution, allowing atomic-level structural analysis. Compositional analysis was carried out using a Super-X EDX detector system, providing an energy resolution of 137 eV for accurate chemical and electronic structure elucidation. To minimize beam-induced damage during transmission electron microscopy characterization, the electron dose was carefully controlled to remain below $300 \text{ e}^- \text{Å}^{-2}$.

DFT calculation

First-principles calculations were carried out using the plane-wave pseudopotential method based on the Perdew–Burke–Ernzerhof exchange–correlation functional⁵⁹, which is implemented using the Vienna *ab initio* simulation package (v. 5.4.4)⁶⁰. The valence wavefunctions were expanded by a plane wave with a cut-off energy of 400 eV, whereas all the structures were optimized until the force on each atom was less than 0.02 eV Å^{-1} . The I/Br and Cs/FA relative distributions in the bulk of perovskites were explored and the model with the lowest energy was chosen to establish the surface models. 2×2 (100), (110) and (111) surface models were established to simulate the different crystal surfaces of perovskites. The vacuum layer is set as 15 Å. The Brillouin zone was sampled using $3 \times 3 \times 1$ *k*-point mesh for geometry optimization. The vacancy formation energy of the FABr pair was calculated according to the following equation:

$$E_{\text{perovskite surface}} = E_{\text{perovskite surface with } V_{\text{FA}} \text{ and } V_{\text{Br}}} + E_{\text{FABr(g)}}.$$

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data supporting this study are included in the article and its Supplementary Information. Further information are available from the corresponding authors on request.

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Acknowledgements

Y.H. acknowledges support from the Agency for Science, Technology and Research (A*STAR) under its MTC IRG Grant (M23M6c0108). We are affiliated with the Solar Energy Research Institute of Singapore (SERIS), a research institute at the National University of Singapore (NUS). SERIS is supported by the NUS, the National Research Foundation Singapore (NRF), the Energy Market Authority of Singapore (EMA) and the Singapore Economic Development Board (EDB). We acknowledge that computational work involved in this research work is fully supported by NUS IT's Research Computing group under grant number NUSREC-HPC-00001. J.A.S. acknowledges financial support from the Australian Research Council (DE230100173). We thank the staff of the BL11 NCD-SWEET beamline at ALBA Synchrotron for their assistance in recording the GIWAXS data.

Author contributions

Z. Dong and Y.H. conceived the idea and designed the experiments. Y.H. directed and supervised the project. Z. Dong fabricated the PSCs and conducted stability tests under accelerated aging. J.H., Y.L. and M. Sui conducted the STEM characterization and data analysis. Z. Dong and X.G. investigated the thermal degradation mechanisms and conducted the material and device characterizations. Z.S. helped with the XRD measurement. Q.Z., T.W. and L.K.L. helped with the device fabrication. J.A.S., Z. Degnan and E.S. performed the *in situ* GIWAXS measurements. H.C. and Yunluo Wang helped with the pole figure measurements. R.L. contributed to the DFT simulations. J.C. helped with the XPS measurements. N.K. and M. Stollerfoht helped with the bias-assisted charge extraction measurement. J.L., N.L. and Yudian Wang helped with the thermal evaporation of perovskite films. Z. Dong and Y.H. analysed the results and composed the paper. All authors discussed and polished the paper.

Competing interests

Y.H. is the founder of Singfilm Solar, a company commercializing perovskite photovoltaics. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-025-02375-8>.

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Peer review information *Nature Materials* thanks Shangfeng Yang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes	0.055 cm2 and 1 cm2.
	☐ No	
Method used to determine the device area	☒ Yes	Corresponding masks were used during the test.
	☐ No	

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	We provide J-V plots in both forward and backward direction in Fig. 4a and Supplementary Fig. 27.
	☐ No	
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes	See details in Method.
	☐ No	
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes	See details in Method.
	☐ No	
Protocol for preconditioning of the device before its characterization	☐ Yes	No preconditioning condition is used.
	☒ No	
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes	We provided MPPT data at different temperatures in Fig.4d, 4e and Supplementary Fig. 24, and stability test results under dark storage in Supplementary Fig. 23.
	☐ No	

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes	Small hysteresis was observed for both of perovskite single junction devices and perovskite-Si tandem devices.
	☐ No	
Related experimental data	☒ Yes	For single-junction, data is provided in Supplementary Table 3. For tandem, data is provided in Supplementary Fig. 27.
	☐ No	

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	We provide EQE measurements in Fig. 4b for perovskite single-junction device and Supplementary Fig. 27 for perovskite-Si tandem.
	☐ No	
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	We compare the integrated Jsc with one from JV scan. The difference between the integrated Jsc from EQE and Jsc from JV scan is about 3% for both of perovskite single-junction and perovskite-Si tandem solar cells, which is within accuracy confidence of the measurements.
	☐ No	
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes	See details in Method.
	☐ No	

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes	See details in Method.
	☐ No	
Confirmation that the reference cell was calibrated and certified	☒ Yes	Our solar simulator was calibrated by Si reference cell (certificated by Newport.)
	☐ No	

Calculation of spectral mismatch between the reference cell and the devices under test

☐ Yes
☒ No

The Jsc from IV scan is within 3% error with respect to integrated Jsc value from EQE.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

Optical aperture masks (0.055 cm² and 1 cm²) were used.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

No significant variations were observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☐ Yes
☒ No

Our paper does not emphasize solar cell efficiency.

A copy of any certificate(s)

Provide in Supplementary Information

☐ Yes
☒ No

Our paper does not emphasize solar cell efficiency.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

12 solar cells are included for statistics of device J-V performance. For every temperature of MPP testing, the normalized PCE curve is from statistics of 3-5 solar cells.

Statistical analysis of the device performance

☒ Yes
☐ No

We provide J-V statistics in Supplementary Table 4 and MPP statistics in Fig. 4e.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes
☐ No

We provided stability data in Fig. 4d, 4e and Supplementary Fig. 23 and 24.